# Hybrid UAV Waypoint Tracking Simulation, Results Anlysis and Optimization

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## Abstract

This document focuses on the simulation results, analysis and corresponding optimizations of "Hybrid UAV Waypoint Tracking" simulated by Panerati et al. [2021].

## 1 Mathematical Model

### 1.1 State Space (in 3D space)

**For *Continuous State*:**

- Position: $p \in \mathbb{R}^3$
- Velocity: $v \in \mathbb{R}^3$

**For *Discrete State*:**

- Index of Waypoint (Tracking Target): $q \in \mathcal{Q} = \{0, 1, 2, \ldots, N\}$
  - Coordinates $W = \{W_0, W_1, \ldots, W_N\}$, where $W_q \in \mathbb{R}^3$ is the coordinate of idx $q$

Thus, the complete **hybrid state vector $\boldsymbol{x}$** of the system is defined as:

$$x = \begin{bmatrix} p \\ v \\ q \end{bmatrix} \in \mathbb{R}^6 \times \mathcal{Q} \tag{1}$$

### 1.2 Flow Set $C$

The flow set determines the conditions under which the system undergoes continuous-time physical evolution:

- As long as the Euclidean distance between the UAV and the current target waypoint $W_q$ is greater than or equal to the tolerance radius $\epsilon$
- or if the system has reached the final waypoint ($q = N$), it maintains continuous flight:

$$C = \left\{ x \in \mathbb{R}^6 \times \mathcal{Q} \mid \|p - W_q\| \geq \epsilon \vee q = N \right\} \tag{2}$$

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### 1.3 Flow Map $f$

Within the flow set, the evolution of the system follows the **differential equation $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x})$**. During this phase:

- For continuous state:
  - $\dot{p} = v$
  - $\dot{v} = \mathcal{F}_{dyn}(p, v, u_{pid}(p, v, W_q))$
- For discrete state $q$: $\dot{q} = 0$ since $q$ is invariant in flow set
- Physical state is governed jointly by the underlying *rigid body dynamics* and the *closed-loop PID controller*:

$$\dot{x} = f(x) = \begin{bmatrix} v \\ \mathcal{F}_{dyn}(p, v, u_{pid}(p, v, W_q)) \\ 0 \end{bmatrix}, \quad x \in C \tag{3}$$

Note:

- $u_{pid}$ is the *control command* with $W_q$ acting as the absolute target
- $\mathcal{F}_{dyn}$ represents the *rigid body dynamics integration* computed by the underlying *6-DOF physics engine*

### 1.4 Jump Set ($D$)

The jump set defines the strict boundary where discrete events (*transients*) are triggered:

- When the UAV enters the $\epsilon$-neighborhood of the target waypoint
- and the current waypoint is not the final destination

The state machine triggers a jump:

$$D = \left\{ x \in \mathbb{R}^6 \times \mathcal{Q} \mid \|p - W_q\| < \epsilon \wedge q < N \right\} \tag{4}$$

### 1.5 Jump Map ($g$) (*i.e. Transition*)

When the state $x \in D$, the system undergoes a discrete jump $x^+ = g(x)$.

- At this instant, the physical position and velocity remain continuous (**no spatial mutation occurs**)
- but the target waypoint index $q$ advances to the next one:

$$x^+ = g(x) = \begin{bmatrix} p \\ v \\ q+1 \end{bmatrix}, \quad x \in D \tag{5}$$

### 1.6 Overall Hybrid Automaton Equations

$$\mathcal{H} := \begin{cases} \dot{x} = f(x), & x \in C \\ x^+ = g(x), & x \in D \end{cases} \tag{6}$$

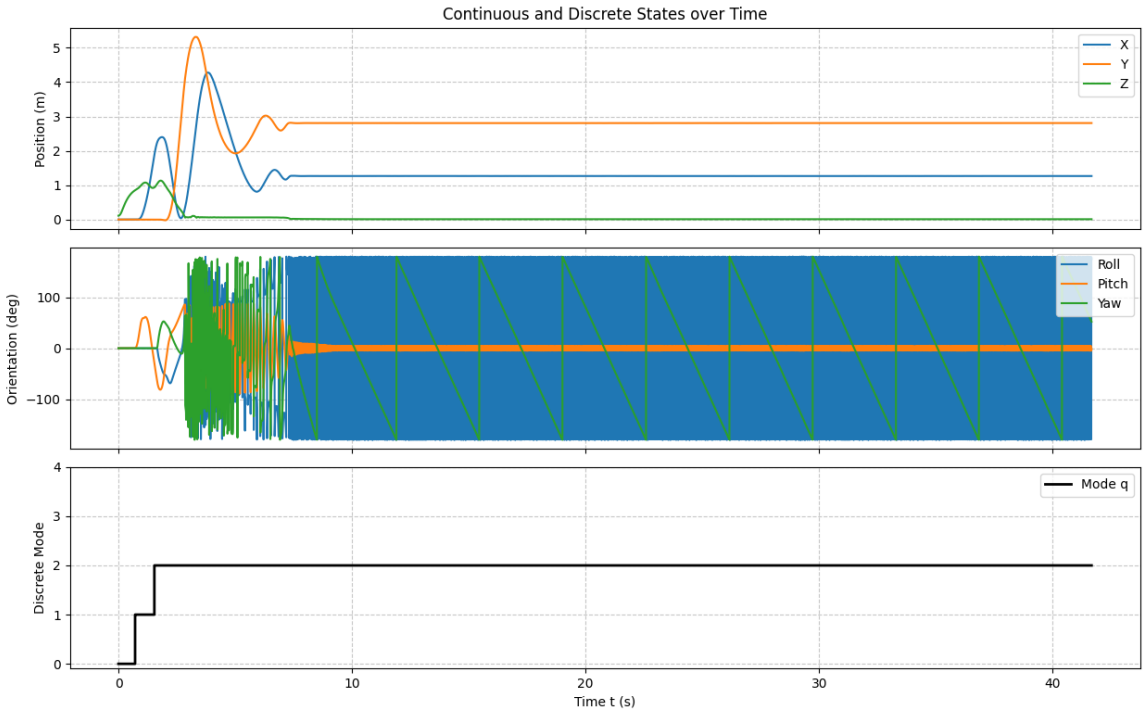


Figure 1: Initial Exp: State Plot

## 2 Simulation Results

### 2.1 Initial Exp

**State & Trajectory Plot** The initial experiment's state plot and trajectory plot see fig. 1 and fig. 2.

### 2.2 Simulation Results Analysis

**Conclusion:** The breakdown of local stability in **underactuated systems** under large step inputs.

**Overview Results Analysis:** As observed in the state and trajectory plots, the UAV successfully tracks the initial waypoints, indicated by the discrete mode transitions ($q = 0 \to 1 \to 2$). However, upon switching to mode $q = 2$ (targeting [2.0, 2.0, 1.5]), the large spatial distance introduces a severe step-error to the closed-loop controller. This forces an aggressive attitude adjustment, evident from the violent high-frequency oscillations in the Orientation plot (Roll/Pitch/Yaw) after $t \approx 2s$. The extreme tilt angles cause the vertical thrust component to drop below gravity ($T \cos\theta < mg$), leading to an irreversible altitude loss ($Z \to 0$, as seen in the Position plot). The UAV hits the ground and becomes locked in an unrecoverable physical state, failing to ever enter the $\epsilon$-neighborhood (Jump Set $D$) of the subsequent waypoint. Consequently, the discrete state $q$ permanently stalls at 2.

**Potential Dynamical Properties to Study:**

- **Stability (Region of Attraction):** The continuous flow map $f(x)$ utilizing standard PID control is only *locally asymptotically stable* for **the 6-DOF underactuated system**. It exhibits severe instability under large step inputs due to the coupling between attitude and translational dynamics.
- **Attractivity (Reachability):** The target jump set $D_{q+1}$ is not attractive from the post-jump initial state $x(t_0)$ under the current flow map. The continuous trajectory diverges to the ground singularity before intersecting $D$.

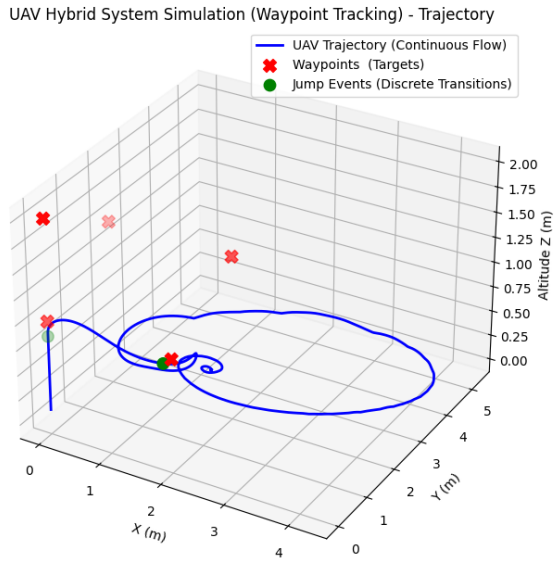


Figure 2: Initial Exp: Trajectory Plot

- **Robustness:** The current system architecture lacks robustness against actuator saturation (motor RPM limits) and unmodeled ground-effect disturbances when subjected to extreme attitude commands.

**Outline of a Possible Analysis Approach:**

- **Step 1: Lyapunov-based Stability Analysis:** Construct a Control Lyapunov Function (CLF), denoted as $V(x)$, for the closed-loop continuous dynamics to estimate the **Maximum Region of Attraction (ROA)** regarding the allowable positional tracking error.
- **Step 2: Forward Reachability Verification:** Mathematically verify whether the initial state immediately following a discrete jump, $x^+ = g(x)$, safely resides within the established ROA of the subsequent flow map.
- **Step 3: Trajectory Redesign (Controller Synthesis):** Introduce a sufficiently smooth reference trajectory planner (e.g., a minimum-snap or cubic polynomial interpolator) into $f(x)$. This ensures the instantaneous tracking error remains strictly bounded within the ROA, guaranteeing forward invariance and successful intersection with all subsequent jump sets.

### 2.3 Optimization Exp

**Optimizations**

- **Virtual Target Initialization:** Introduced a *virtual_target_pos* state variable initialized exactly at the drone's spawn location:
  - This ensures the **initial control error is zero**,
  - **preventing any violent takeoff maneuvers**.

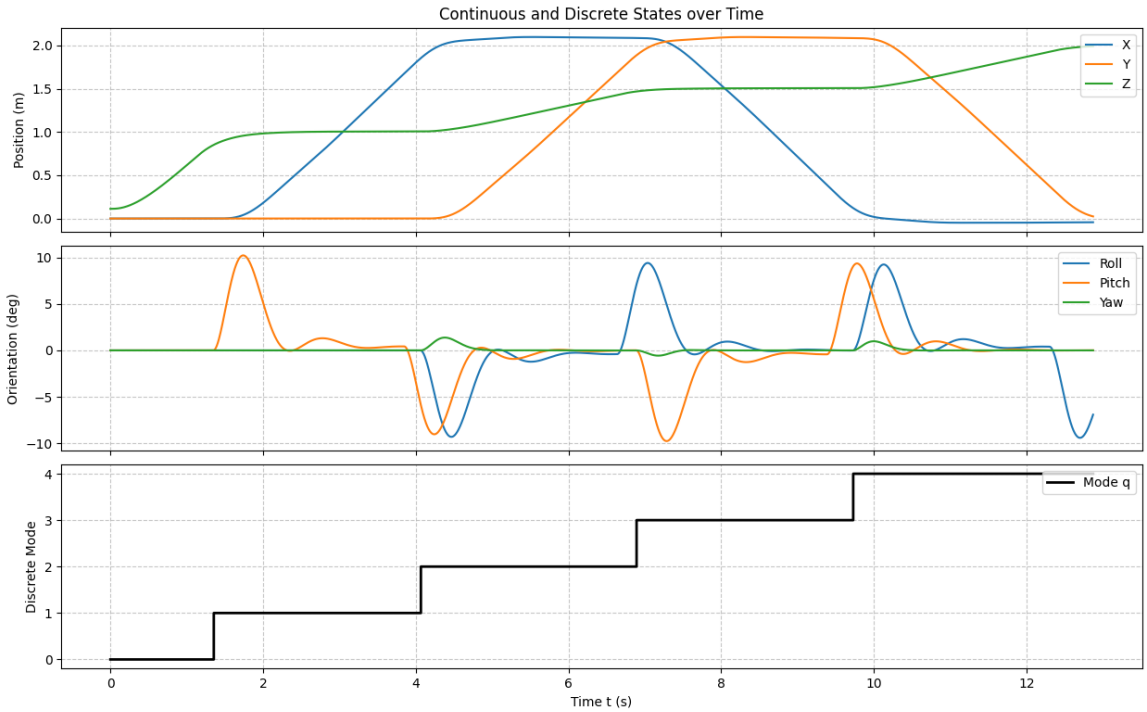


Figure 3: Optimization Exp: State Plot

- **Kinematic Trajectory Generator (Linear Interpolation):** Added a **smooth interpolation mechanism** *inside* the continuous Flow Map $f(x)$ loop. Instead of instantly switching to the next distant waypoint, the virtual target moves towards the absolute waypoint at a **constrained, constant velocity** (MAX_SPEED = 0.8 m/s). -> Enhance Stability
- **Elimination of Step-Inputs for the Controller:**
  - Modify the PID controller to track this continuously moving *virtual_target_pos* rather than the absolute *WAYPOINTS[q]*.
  - By breaking down the large spatial gap into millimeter-level, time-stepped increments, we completely eliminated the massive step-errors that previously caused extreme attitude (pitch/roll/yaw) oscillations and the local stability failure of the underactuated system.

**State & Trajectory Plot & GUI Display** The optimization experiment's state plot, trajectory plot and gui display see fig. 3, fig. 4 and fig. 5.

## References

J. Panerati, H. Zheng, S. Zhou, J. Xu, A. Prorok, and A. P. Schoellig. Learning to fly—a gym environment with pybullet physics for reinforcement learning of multi-agent quadcopter control. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 7512–7519, 2021. doi: 10.1109/IROS51168.2021.9635857.

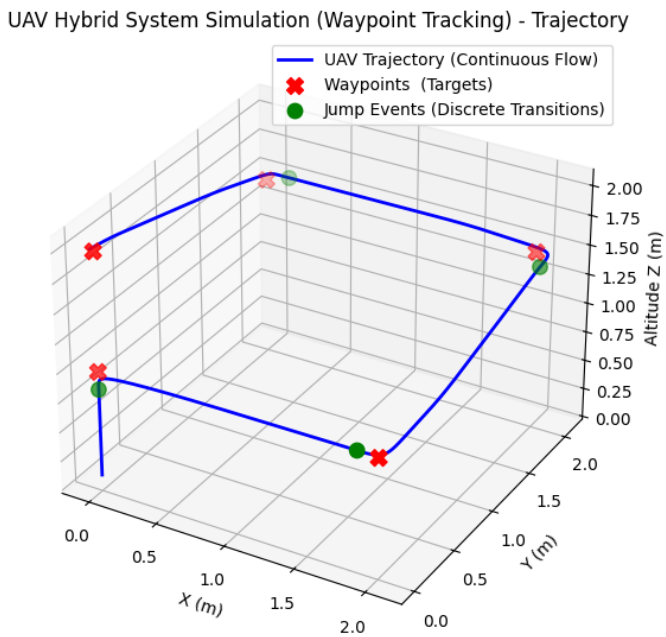


Figure 4: Optimization Exp: Trajectory Plot

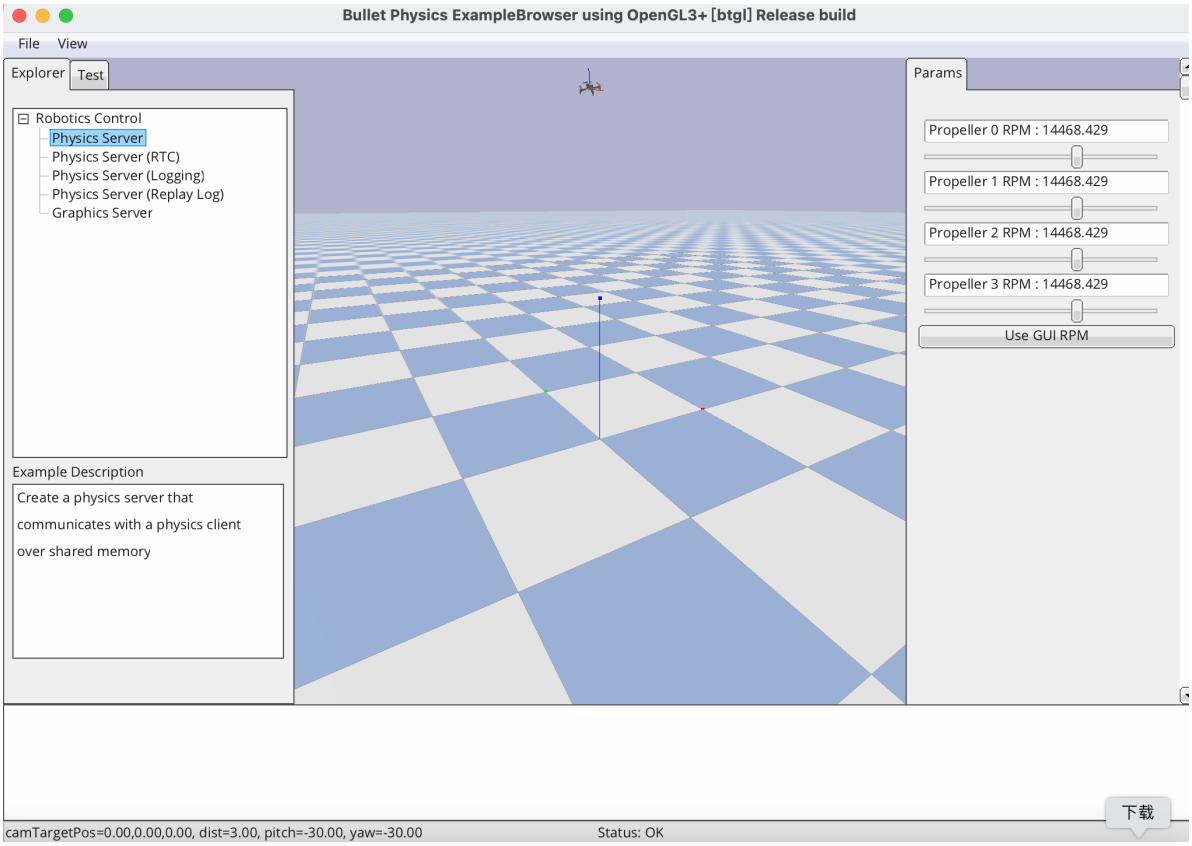


Figure 5: Optimization Exp: Simulation GUI Display